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in
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AI-based object detection on point clouds for privacy awareness

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Abstract

Περίληψη

Acknowledgements

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place.

If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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1 Introduction

1.1 Motivation and Problem Statement

1.2 Thesis Structure

Chapter 2

Chapter 3

Chapter 4

Chapter 5

Table 1.1 refers to the results in Figure 4.1.

	Name	Task	Type
1	Tom	Seek	Cat
2	Jerry	Hide	Mouse

Tab. 1.1: This is a test table

Background and Related Work

2

2.1 Background

2.1.1 Background Information and Terminology

Point clouds are a set of points positioned in three-dimensional space in the form of (x,y,z) coordinates, where z is the depth within the two-dimensional image. Depending on the use case and the capture device points can be associated with other data such as colors and reflectance properties [26a]. Point clouds are captured using specialized devices such as LiDAR(Light Detection and Ranging) cameras that project laser pulses and measure return time in order to provide the third dimension of the points [26b]. There are other technologies for achieving this result but are out of the scope of this work. LiDAR cameras are mostly utilised in cars for achieving self-driving capabilities. Other scenarios are outside of the scope of this work.

The most popular file formats for point clouds are .PLY which is a polygon file format (binary and ascii text variants) and .LAS for LiDAR data. This is where i can mention the intel realsense contribution.

Object detection is a domain of computer vision that attempts to classify objects within an image and find their bounds. In the case of 2D object detection it is about providing a rectangle and in Point Cloud (PC)s it is about providing a cube.

Face detection is a specific case of object detection that is useful in different scenarios. Not to be confused with face recognition, which is about comparing a face scan against the scans from a face recognition database [SBJ17]. In that domain there is more focus about extracting facial features and trying to discern different faces. In this work we only explore object and face detection and not face recognition.

Privacy is about X(probs from wikipedia, does not need a formal reference).

Point clouds produced by LiDAR cameras are around 100k points per frame, a number changing based on the technical specification of the camera. In some cases point clouds are converted into a three-dimensional grid comprised of cubes that each contains a certain

number of points. These cubes are also known as *voxels* in the literature and are primarily used for reducing the computational cost of operations on PCs and grouping neighboring points.

Add some extra images.

2.1.2 Object detection in point clouds

Object detection in PCs imposes extra constraints on the AI architectures developed for learning based tasks. The model result should ideally follow some prescriptive rules in order to be effective at the task at hand as outlined in [Qi+17a]. Those rules are implicitly followed in all methodologies presented in this section.

1. The ordering of the points should not influence the end result.

2. Transformations of the points, such as translation(moving the points around together) and rotation should not change the final prediction.

3. The neighborhood of a point contains significant information that should be actively utilised in the learning process.

The model should be able to handle sparse point clouds, as they commonly occur when doing captures using LiDAR cameras.

Foundational work in object detection for PCs has been done in [Qi+17a]. In this work the authors introduce PointNet, a neural network architecture that operates on PCs and not some other derivative form as we will see in other approaches. They approximate a symmetric function by utilising a multi-layer perceptron (MLP) and a max-pooling function and therefore achieving the first model requirement as outlined earlier. The work in this model and the refined version in the next paragraph is reused in newer architectures.

The authors of PointNet followed up on their work and presented [Qi+17b] where they introduce an enhanced version of PointNet, called PointNet++. In this work they replaced the single max pooling operation of PointNet with a hierarchical mechanism that samples centroids from the point cloud, groups the points in regions defined by the centroids and then extracts local features from the regions. In this way, patterns are captured in multiple scales making features more detailed and robust.

In [ZT18] the authors take a different approach and instead of raw-point clouds their model, VoxelNet, uses voxels as the data representation, allowing them to take advantage of more points in a point cloud efficiently. More specifically they introduce a voxel feature

encoding layer(VFE) layer that summarizes point-wise features in a voxel. Stacking VFE layers achieves a high-dimensional representation, which is then used by a region proposal network(RPN) for object detection.

In [SWL19] the authors propose a 2 stage object detection methodology based on R-CNNs and PointNet++ as a backbone network. The first stage generates a small number of 3D bounding boxes and separates points into foreground and background ones. The second stage then refines those proposals using the information produced in the first stage.

The work in [SR20] brings the graph neural network(GNN) architecture into point clouds. It constructs a graph of the point cloud by adding edges between neighboring points in a fixed radius. For the detection it uses a MLP for classification and a MLP that computes a bounding box for each class.

the work in [Pan+21] introduces PointFormer an architecture that does those things.

The work in [YZK21] introduces CenterPoint, an architecture based on the X,Y,Z concepts.

Another work that uses the Voxel-based approach is [Den+21] where the authors utilize the R-CNN architecture for learning tasks on point clouds.

2.1.3 Person/Face detection in point clouds

It is basically about taking existing models and training them on more specific datasets and changing slightly the architecture to accomodate the specific use case. In [JL21] the authors focus on identifying people for LiDAR point clouds.

2.1.4 Datasets for point clouds object detection

Some information regarding the general form of the datasets. For face detection extra processes are required to get access to the datasets. Datasets containing faces are mostly for tasks such as face recognition, pose estimation and other tasks that require more tuning than conventional object detection. They differ significantly from object detection datasets as they already have cropped and aligned the faces in the data without leaving the rest of the background intact.

Most datasets for object detection in point clouds are for scenarios involving self-driving vehicles and are captured by LiDAR devices. They have pedestrians as a category but not faces.

In [Gei+13] they present the kitti dataset. In [Yu+20]the dataset contains... In [Mao+21] they present the once dataset In [Wil+23] they present the dataset ArgoV2 where...

2.2 Related Work

[Ull+22] Commercial software found at: `https://www.riegl.com` and at `https://www.celantur.com/`.

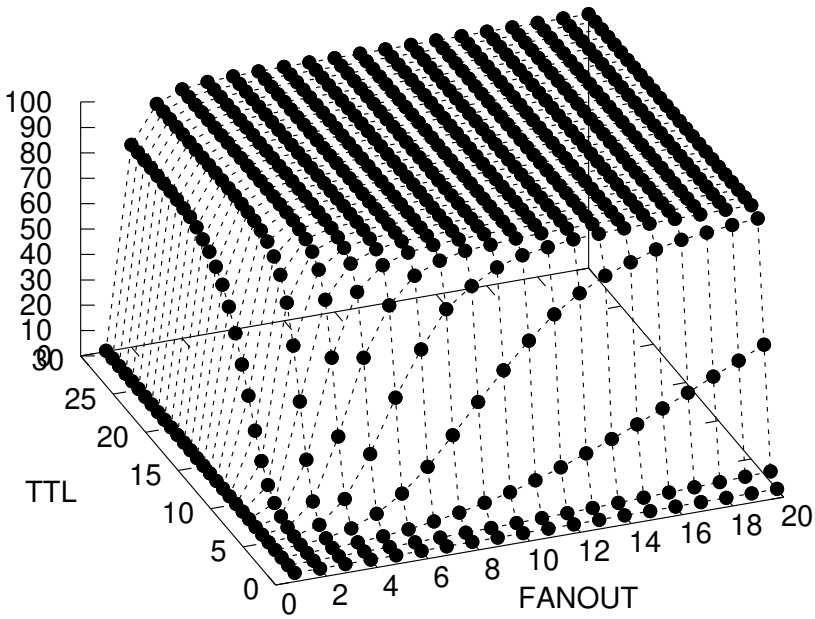


Fig. 2.1: Caption...

3 System Design and Implementation

3.1 Design

- Item 1
- Item 2
- Item 3
- Item 4

3.2 Implementation

Algorithm 1 Test

```
procedure GET(a, b)                    ▷ The g.c.d. of a and b
    r ← a mod b
    while r ≠ 0 do                     ▷ We have the answer if r is 0
        a ← b
        b ← r
        r ← a mod b
    return b                           ▷ The gcd is b
```

4 Evaluation

4.1 Experimental setup

4.2 System evaluation

In this section we present the performance evaluation of the proposed system.

4.2.1 Performance and cost evaluation

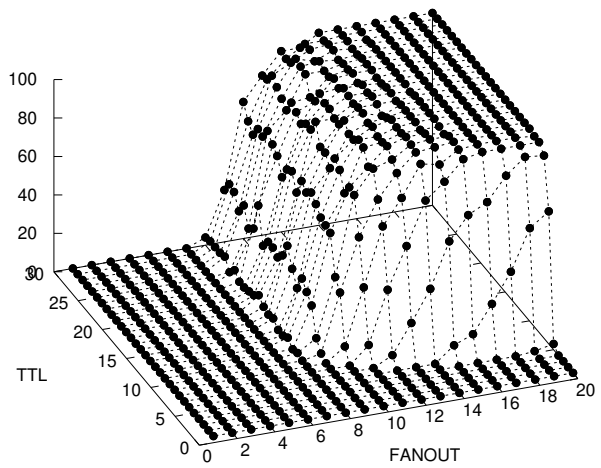
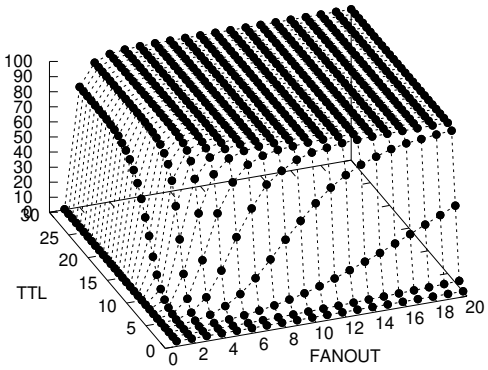


Fig. 4.1: This figure shows the percentage of complete disseminations.

4.2.2 Qualitative evaluation

First column	Second column	Third column
1	2	3
4	5	6
7	8	9

Tab. 4.1: Test table



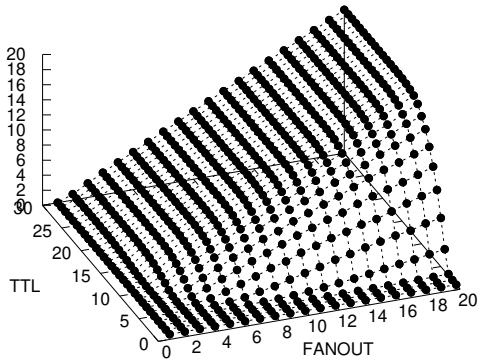


Fig. 4.2: These are the remaining two figures.

4.2.3 Effect of other parameters

Conclusions 5

5.1 Limitations

The 3D Object Detection python ecosystem is not that mature yet. There are great but unmaintained projects unsuitable for software in production environments. More specifically, python versions are outdated, only specific versions of packages are supported. Pretrained and actually useful models are rare in contrast to 2D where for example opencv offers out of the box several object and face detection modules.

Face detection as a specific task is not that popular in Point cloud related settings. Mostly pedestrians in an outdoor setting are actually the target of the latest models. There has been more focus in face recognition tasks as point clouds can provide richer features of a face than the two-dimensional equivalent and can therefore improve the effectiveness of face recognition systems.

Most of the attention is on traffic settings, other use cases remain mostly unexplored e.g. indoors 3D cameras. In [Wan+22] the authors improve upon an existing model to make it more suitable for complex traffic conditions.

5.2 Discussion

5.3 Future Work

Bibliography

[26a] en. Page Version ID: 1341468063. Mar. 2026.

[26b] en. Page Version ID: 1337706693. Feb. 2026.

[Den+21] Jiajun Deng, Shaoshuai Shi, Peiwei Li, et al. "Voxel R-CNN: Towards High Performance Voxel-based 3D Object Detection". en. In: *Proceedings of the AAAI Conference on Artificial Intelligence* 35.2 (May 2021), pp. 1201–1209.

[Gei+13] A Geiger, P Lenz, C Stiller, and R Urtasun. "Vision meets robotics: The KITTI dataset". en. In: *The International Journal of Robotics Research* 32.11 (Sept. 2013), pp. 1231–1237.

[JL21] Dan Jia and Bastian Leibe. *Person-MinkUNet: 3D Person Detection with LiDAR Point Cloud*. arXiv:2107.06780 [cs]. July 2021.

[Mao+21] Jiageng Mao, Minzhe Niu, Chenhan Jiang, et al. *One Million Scenes for Autonomous Driving: ONCE Dataset*. arXiv:2106.11037 [cs]. Oct. 2021.

[Pan+21] Xuran Pan, Zhuofan Xia, Shiji Song, Li Erran Li, and Gao Huang. "3d object detection with pointformer". In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2021, pp. 7463–7472.

[Qi+17a] Charles R. Qi, Hao Su, Kaichun Mo, and Leonidas J. Guibas. "Pointnet: Deep learning on point sets for 3d classification and segmentation". In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017, pp. 652–660.

[Qi+17b] Charles Ruizhongtai Qi, Li Yi, Hao Su, and Leonidas J. Guibas. "Pointnet++: Deep hierarchical feature learning on point sets in a metric space". In: *Advances in neural information processing systems* 30 (2017).

[SBJ17] Sima Soltanpour, Boubakeur Boufama, and Q. M. Jonathan Wu. "A survey of local feature methods for 3D face recognition". In: *Pattern Recognition* 72 (Dec. 2017), pp. 391–406.

[SR20] Weijing Shi and Raj Rajkumar. "Point-gnn: Graph neural network for 3d object detection in a point cloud". In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020, pp. 1711–1719.

[SWL19] Shaoshuai Shi, Xiaogang Wang, and Hongsheng Li. “Pointrcnn: 3d object proposal generation and detection from point cloud”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition.* 2019, pp. 770–779.

[Ull+22] Rehmat Ullah, Hassan Hayat, Afsah Abid Siddiqui, et al. “A Real-Time Framework for Human Face Detection and Recognition in CCTV Images”. en. In: *Mathematical Problems in Engineering* 2022 (Mar. 2022). Ed. by Nouman Ali, pp. 1–12.

[Wan+22] Hai Wang, Zhiyu Chen, Yingfeng Cai, et al. “Voxel-RCNN-Complex: An Effective 3-D Point Cloud Object Detector for Complex Traffic Conditions”. In: *IEEE Transactions on Instrumentation and Measurement* 71 (2022), pp. 1–12.

[Wil+23] Benjamin Wilson, William Qi, Tanmay Agarwal, et al. *Argoverse 2: Next Generation Datasets for Self-Driving Perception and Forecasting.* arXiv:2301.00493 [cs]. Jan. 2023.

[Yu+20] Fisher Yu, Haofeng Chen, Xin Wang, et al. “Bdd100k: A diverse driving dataset for heterogeneous multitask learning”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition.* 2020, pp. 2636–2645.

[YZK21] Tianwei Yin, Xingyi Zhou, and Philipp Krahenbuhl. “Center-based 3d object detection and tracking”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition.* 2021, pp. 11784–11793.

[ZT18] Yin Zhou and Oncel Tuzel. “Voxelnet: End-to-end learning for point cloud based 3d object detection”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition.* 2018, pp. 4490–4499.

List of Acronyms

API Application Programming Interface

WWW World Wide Web

PC Point Cloud

MLP multi-layer perceptron

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4.2 These are the remaining two figures. 10

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